

Institute of Actuaries of India

Subject SA4: Pension & Other Employee Benefits

Examination Diet: May 2013

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the paper setters with the aim of helping markers of scripts so as to have a framework and be consistent while evaluating answers. The solutions given are only indicative. It is realized that there could be other points as valid answers and the marker may give credit for any such alternative approach or interpretation which the marker considers to be appropriate.

- Q. 1)** A large manufacturing firm based in Germany is looking to acquire an Indian manufacturing company. The target is an old Indian company providing multiple employee benefits besides the mandatory Gratuity and Provident fund benefits. The acquirer has a small legal entity in India and the target, if successfully acquired, would fold into the existing legal set up.

The acquiring company is concerned about the risks that may be underlying these benefit liabilities and hires your firm to assist with the due diligence. You are a qualified Actuary and hold a senior Consulting position within your firm.

From the data room information you understand that the following benefits are being provided by the Indian Company being acquired.

1. Gratuity
2. Provident fund
3. Leave encashment
4. Sick leave
5. Post-retirement medical benefit (PRMB) &
6. Defined benefit pension

The following liability information besides copies of actuarial certificate is available in the data room;

1. Number of employees covered by the Gratuity and Leave encashment benefits : 2,200
2. Gratuity liability as per 31 March 2013 Valuation: INR 45 crores
3. Leave encashment liability as per 31 March 2013 Valuation: INR 25 crores
4. Number of employees covered by the Post-retirement medical benefit and Defined benefit pension is 800 as only the unionized employees are eligible for these benefits
5. PRMB liability as per 31 March 2013 Valuation: INR 18 crores
6. DB pension liability as per 31 March 2013 Valuation: INR 20 crores

On examination of the actuarial certificates, it is found that the following bases have been used for valuations as at 31 March 2013

1. Salary escalation: 5%
2. Medical inflation for the PRMB benefit : 4%
3. Mortality : LIC 94-96 modified Ultimate
4. Attrition : 4% across all ages

Also the benefit provisions are as below;

1. **Gratuity:** as per Act provisions for service less than or equal to 10 years and allows higher accrual at 21 days for service greater than 10 years with no ceiling on benefit payable. Gratuity is funded and the funding level is 75% as at 31 March 2013
2. **Leave:** annual entitlement at 30 days, maximum accumulation – 90 days, encashment formula – leave accumulation * leave salary/26
3. **Sick leave:** annual entitlement at 12 days, maximum accumulation – 45 days, encashment formula same as Privilege leave
4. **PRMB:** hospitalization cover of INR 500,000 to member and spouse for life post retirement and payable only on retirement
5. **DB Pension:** INR 1,000 for every year of service subject to a maximum of 50% of last drawn pay payable for life only on retirement

The existing legal entity in India of the acquirer provides only the mandatory Gratuity and Provident fund benefits.

You have been asked to provide:

- i) A due diligence report addressed to the management of the acquirer covering;
 - a. The nature and extent of the risks under each of the defined benefit schemes demonstrated with sample calculations in the context of the benefit liabilities along with commentary on the current assumptions that has been used for the valuations. (20)
 - b. Data that would be required and analysis that needs to be done to validate the current assumptions being used (5)
 - c. Significant differences between Indian and IFRS accounting for the year ending 31 March 2013. (5)

- ii) The acquirer would also want to understand the options for transferring the defined benefit schemes and financial implications. Provide at least three options for transferring each benefit provided by the target into the existing Indian legal entity of the acquirer and the pros and cons of each option from the perspective of the target company, the employees concerned, the acquirer and the legal entity of the acquirer (20)

Solution 1(i)

- a) We understand that Company XYZ based in Germany is looking to acquire an Indian manufacturing company. The acquiring company would like to understand the nature and extent of risks of the Defined benefit schemes provided by the Indian Company and this report has been prepared to provide a comprehensive understanding of the nature and extent of risks under the following plans namely

1. Gratuity
2. Provident fund
3. Leave encashment
4. Sick leave
5. Post-retirement medical benefit (PRMB) &
6. Defined benefit pension

In addition, the report also covers the data required to validate the assumptions being used for the valuations and the significant differences between IFRS and Indian GAAP reporting.

Nature and extent of risks under the different defined benefit aspects of the six plans:

Gratuity:

All the benefits under Gratuity scheme are in the nature of Defined Benefits. Gratuity Act applies to nearly all employers who employ (or had employed on any day preceding 12 months) ten or more employees. It becomes obligatory on the part of such employers to pay gratuity to their employees. Hence Gratuity payable as per the Gratuity Act (Act Gratuity) is a statutory benefit for all employees irrespective of their salary.

As per Gratuity Act, the gratuity is computed at the rate of 15 days' wages for each year of service. For this purpose a month's wages are treated as 26 days' wages and hence the benefit accrues at the rate of 15/26 of monthly wages for each year of service. The Act gratuity is thus calculated as under:

[50]

Act Gratuity = $15/26 \times \text{Service in years} \times \text{monthly wages}$

A part of service in excess of six months is taken as one year and wages are the last drawn monthly wages of the employee concerned. Gratuity is payable immediately on the “termination” of employment of the employee after he has rendered a continuous service of not less than five years:

- on his superannuation; or
- on his retirement; or resignation; or
- on his death; or disablement due to accident or disease.

The completion of continuous service of five years is not necessary in case of death/disablement.

As per the Act, the amount of gratuity shall not exceed INR 1 Million. The Gratuity Act, however, does not deny the right of an employee to receive better terms of gratuity under any award or agreement or contract with the employer. An employer may, therefore, pay gratuity at a rate of more than fifteen days' wages (e.g. one month wages) for each year of service with or without any ceiling. The Act merely prescribes a minimum gratuity and therefore no employer can provide less than the Act benefit.

In the instant case the Gratuity benefits provided by the Company is better than the Act provisions. The higher accrual in later years of service coupled with no ceiling on the benefit payable would result in material increases in liability as the average service of the employee population increases.

The assumption of salary escalation of 5% p.a. is low given that the long term price inflation in India has been in excess of 5% and hence the liability is probably underestimated unless there are strong reasons to justify the assumption used.

The impact on the liability due to a one percent revision in the salary escalation assumption would be material as the liability is uncapped. Also the percentage impact would depend on the average future working life time of the employee population that will be based on the average age, retirement age and attrition assumption used.

The funding level of 75% would also reduce depending on the increase in the liability value due to the impact of a suitable increase in the salary escalation assumption.

Provident fund:

Only in a situation where the Provident Fund is an 'except' scheme (not clear in the question), there is an element of defined benefit otherwise the scheme is defined contribution. Generally, Provident Fund is a defined contribution benefit where contributions are made by both the employer and the employees in equal proportion with optional additional contribution by the employees during the working lifetime of employees. The accumulated value as a lump sum is available to the employee on leaving service.

Assuming that the benefit is administered by RPFC, there would be no risks arising owing to the

liability given that the scheme is defined contribution in nature.

Alternatively if an assumption is made that the provident fund is self-managed and there is a risk owing to the liability to the Company on account of the interest guarantee on the fund, the volatility and unpredictability of the returns on RPFC.

Leave encashment: (Privilege leave and Sick leave)

In the instant case the benefits are in the nature of defined benefits. It is common practice to provide leave encashment facility to employees. Such facility is generally provided for un-availed Earned (or Privilege) leave. Some employers, however, provide encashment of Medical (or Sick) leave also. Employees earn and accumulate leave during their service period. There are rules around annual entitlement, maximum carry forward out of annual entitlement, encashment basis, eligibility and ceiling. As per AS-15 (Revised), the employers are required to ascertain accrued liability in respect of accumulated leave (privilege and sick) that is payable in the future at the time of separation on an appropriate basis and to provide for the same in the Statement of Profit & Loss of the Company.

The Company allows for a maximum accumulation of 90 days under privilege leave and 45 days under sick leave. This privilege leave encashment benefit is material as it provides three months' salary at the time of exit compared to a nil benefit in the acquiring entity. Likewise encashment of sick leave is not a prevalent market practice.

The assumption of salary escalation of 5% p.a. is low given that the long term price inflation in India has been in excess of 5% and hence the liability is probably underestimated unless there are strong reasons to justify the assumption used.

The impact on the liability due to a one percent revision in the salary escalation assumption may be material. However, the impact on the purchasing accounting liability would depend on the current accumulated leave balances and if and by how much they are in excess of the maximum accumulation of ninety days and forty five days respectively besides the average future working life time of the employee population which will be based on the average age, retirement age and attrition assumption used.

The scheme is entirely unfunded and hence the increase in the liability value will have a direct impact on the balance sheet due to the increase on account of the salary escalation assumption.

Post-retirement medical benefit:

In the instant case if the benefits under the scheme are insured with insurer, then the nature of liability becomes that of defined contribution otherwise it is defined benefit. The company provides hospitalization cover to employee and spouse for life post retirement capped at INR 500,000 per annum and the benefit is payable only in cases of exit on account of retirement. Typically such cover if bought with an insurance company (Mediclaime insurance offered by general insurance companies) would provide for hospitalization expenses which include

room charges, doctors' fees, operation theatre charges, ICU, drugs and medicines, diagnostic tests etc. pre and post hospitalization expenses are also covered up to certain number of days. Usually these covers are provided with certain exclusions and there are waiting periods for some covers which may be different. If self-managed the terms of the benefit on the above terms must be clearly defined and applied.

Providing a post-retirement medical benefit is not a prevalent practice and has material risks given the defined benefit nature of the benefit.

The benefit is provided to the union grade and is payable only on retirement. There is no other eligibility condition that is defined as minimum service though the impact is likely to be very minimal as the benefit is payable on retirement only.

The costs for providing a 500,000 cover would continue increasing with medical inflation that has been quite high and well in excess of price inflation in the past.

There is a longevity risk as the benefit is payable for the joint life post retirement. With mortality improvements, the duration for which the benefit is due will increase and will have a material impact on the liability.

The current cost basis and the medical inflation used are critical parameters for the valuation. The medical inflation basis of 4% is quite low and lower than long term price inflation while in practice the medical inflation has well been in excess of medical inflation.

The impact on the liability due to a one percent revision in the medical inflation assumption would be material as the cost for providing the cover is uncapped. Also the percentage impact would depend on the current average age and marital status, average future working life time of the employee population that will be based on the average age, retirement age and attrition assumption used.

Defined benefit Pension:

Pension is not a mandatory benefit and defined benefit pension is also not a prevalent benefit in the market. The scheme provided by the target is quite generous as it is applicable to all employees and there are no other eligibility conditions except that the benefit is payable only on retirement.

Assuming that the pension benefit is purchased from an insurer, the scheme is subject to annuity rate risk as the change in purchase basis would impact the liability to the company.

The annuity rate would in turn depend on the mortality improvements and life expectancy, expenses, investment performance and profit margin of the insurance company besides the competitive nature of the annuity market. The rates have been quite volatile in the past. The percentage impact of the change in annuity rate basis would depend on the number of eligible pensioners estimated in the future, the amount of estimated pension payable and the time to payment.

Assuming the scheme is funded, it is also subject to investment risk as the scheme is defined benefit in nature and the additional liability due to poor investment performance would have to be borne by the Company.

The assumption of salary escalation of 5% p.a. is low given that the long term price inflation in India has been in excess of 5% and hence the liability is probably underestimated unless there are strong reasons to justify the assumption used.

The impact on the liability due to a one percent revision in the salary escalation assumption may be material. However the percentage impact would depend on the current employee profile (age, salary level, average past service) and the average future working life time of the employee population which will be based on the average age, retirement age and attrition assumption

(20 Marks)

b. Data that would be required and analysis that needs to be done to validate the current assumptions being used :-

The following assumptions are critical for the purpose of validating the different benefit liabilities;

1. Discount rate and salary escalation assumption (relevant for all the defined benefit schemes provided by the target)
2. attrition assumption (PRMB & DB Pension to a greater extent as eligibility is dependent on the number of people who are estimated to retire; Leave and Gratuity to a lesser extent)
3. medical inflation assumption (for PRMB)

The following data would be required for the purpose of validating the current assumptions being used :-

1. past experience for an adequate period of the scheme members that is complete, appropriate, consistent and of the order of detail required
2. the current HR plan and any communication to employees that has set expectations
3. market practice (general industry and manufacturing industry)
4. Macroeconomic situation (long term inflation, current yields, political scenario etc..)
5. Information on metrics by which the experience may be different (for instance different salary escalation for different cadres, age, service bands etc..)
6. Past assumptions used and any change

The following analysis needs to be done to validate the assumptions being used;

1. Check there is no missing data. For instance new entrants are not missed out.
2. Check the data is appropriate. For instance the salaries provided are the benefit salaries
3. Review high and low salaries and check these with the company if necessary
4. Check that percentage salary increase and ratio of gross to basic salary looks reasonable
5. Check if there were any abnormal events in the past or particular years with special economic events, for ex 2009 was a year of recession and the salary increases in that

year may not be a suitable reference, that needs to be excluded for the purpose of validation.

6. Check if the data is adequate to justify the assumptions basis.
7. Calculate the actual experience for the different assumptions from the past data after necessary checks and identify any trends (general trend, by relevant metrics – grade, age, service etc.

(5 Marks)

c. Significant differences between Indian and IFRS accounting for the year ending 31 March 2013 as the actuarial certificates are as per Indian GAAP(5)

1. The target is currently reporting as per Indian GAAP only while the acquirer has a reporting requirement as per IFRS and would like to understand the purchase accounting implications due to the use of IFRS.

2. The following are the major differences between Indian GAAP and IFRS revised

a. Under Indian GAAP the gains/ losses are recognized directly in the P&L while under IFRS revised, the gains/ losses flow to other comprehensive income.

b. The basis for determination of financing cost is different between the two accounting standards. Under IFRS the net interest cost is determined based on the Interest cost and expected return on plan assets (EROA) replaced with net interest on net DB liability/asset. Hence the discount rate is used instead of the EROA.

c. for Plan changes, under AS15, only the vested past service cost is recognized immediately and the unvested portion is recognized over the period of vesting. However under IFRS revised, the entire past service cost is recognized in P&L.

d. IFRS revised has expanded disclosures as compared to AS15. The plan needs information on sensitivity of liability to key assumptions, commentary on asset liability matching strategies, information on future cash flows providing future expected contributions and benefit payments.

e. IFRS also presents the employee benefit expense in a slightly different format as compared to AS15 taking into account the service cost, financing cost and re-measurement components.

(5 Marks)

Solution 1 (ii)

The choice of method and assumptions to determine the transfer will be negotiated in the context of the overall deal.

It is also necessary to determine the benefits that will be paid to members who transfer to the acquiring entity. This would depend on the following;

- a. Intent of the deal (asset purchase, talent etc...)
- b. benefit philosophy of the acquiring company
- c. the terms of the deal
- d. legislative requirements
- e. union agreements
- f. employment contract
- g. mandatory provisions
- h. the level of funding and transfer of funds
- i. tax status
- j. alignment with existing benefits
- k. retention of key talent
- l. market practice
- m. administrative requirements
- n. Bargaining power of the target
- o. Financial implications and materiality
- p. The current overall compensation structure and market alignment
- q. Need for employee consent
- r. Continuity of service &
- s. Any other valid basis

Options for transferring the benefit:

Gratuity:

The following options could be considered depending on the above aspects :-

- 1. Continue the existing benefits to the transferring employees
- 2. Cash out the additional benefit and provide as per the mandatory act provisions
- 3. Pay out the termination gratuity and with no continuity of service and Gratuity as per act in the acquiring entity

Pros:

- 1. From an employee perspective continuing the existing benefits would be the most favorable and hence employee relations would be good under option one
- 2. From a financial implication perspective option three would be the best as there is no service continuity and benefit is lower than paid currently
- 3. Option two is a via media between options one and two

Cons:

1. Option one would create inequity between existing employees and the transferring employees and may create dissatisfaction and employee issues
2. Option two would create cash flow issues as the cash out would require immediate pay outs assuming the cash out is paid out

Provident fund:

The benefit is mandatory and would be transferred as is into the acquiring company. however the mode of management of the funds (self-managed or RPFC) needs to be evaluated to suitably consider the transfer basis and implications

Leave encashment (privilege & sick leave)

The following options could be considered depending on the above aspects :-

1. Continue the existing benefits to the transferring employees
2. Freeze the benefit and no further accumulation allowed after transfer and as per existing rules in the acquiring company
3. Pay out the leave encashment amount and with no continuity of service and transfer of leave benefit in the acquiring entity

Pros:

1. From an employee perspective continuing the existing benefits would be the most favorable and hence employee relations would be good under option one
2. From a financial implication perspective option three would be the best as there is no service continuity and no encashment benefit is payable
3. Option two is a via media between options one and two

Cons:

1. Option one would create inequity between existing employees and the transferring employees and may create dissatisfaction and employee issues
2. Option three would create cash flow issues as the cash out would require immediate pay outs assuming the cash out is paid out

Post-retirement medical benefit and DB pension

The following options could be considered depending on the above aspects :-

1. Continue the existing benefits to the transferring employees
2. Incorporate additional eligibility conditions, for instance minimum service or payable only for the next five years etc..
3. Cash out an estimated value of the benefit accrued and pay out on termination in future or immediately.

Pros:

1. From an employee perspective continuing the existing benefits would be the most favorable and hence employee relations would be good under option one
2. Option three would be aligned to the benefit philosophy of the acquiring company as it does not provide PRMB/ DB pension to its existing employees
3. Cashing out would reduce the volatility of the financial impact in the future and hence is a favorable one

Cons:

1. Option one would create inequity between existing employees and the transferring employees and may create dissatisfaction and employee issues
2. Option three would create cash flow issues if the cash out would require immediate pay outs
3. Option two may create union issues
4. Option two and three may create administrative issues and complexity
5. Option three may have regulatory constraints etc..

(20 Marks)**[Total Marks-50]**

- Q. 2)** A large multinational company in India is currently reviewing its compensation and benefits policy to be in line with global practices. The company has appointed you as the Consulting Actuary for this review. You have presented your report to the management and recommended introducing a '*Flexible Benefit Scheme*' where the employee could choose the type and level of benefit for a total '*Annual Benefit Amount*'. The program will cover both Statutory and Other benefits of the company including Provident Fund, Gratuity, Leave, Life Insurance and Health Insurance.
- i) Describe briefly the concept of '*Flexible Benefit Scheme*' and how it could be structured. (4)
- ii) Suggest with a reason four benefits which the employees must compulsorily choose. (4)
- iii) You have also included in your report how the company should price for each benefit, which need to be deducted from the '*Annual Benefit Amount*' in case a particular benefit is chosen.
- List the factors which you might have considered in arriving at such a price. (6)
- iv) Explain the Black's Model for valuing interest rate guarantees in exempt provident funds. (8)
- v) The company operates a gratuity benefit scheme as per the Payment of Gratuity Act, 1972 ('*act gratuity*'). The plan has been funded with an insurance company. The fund is in surplus as per the insurance company's latest actuarial valuation. (4)

List the tax benefits to the employer and the employee for a funded gratuity benefit scheme.

- vi) The company has suggested the contribution rate currently paid to the insurance company should be used as the price of the gratuity benefit in the '*Flexible Benefit Scheme*'. (6)
Comment on the suitability of using the current contribution rate as the price.

- vii) List two alternative methods of pricing for the gratuity benefit and their suitability. (2)

- viii) Briefly explain the choice of assumptions as covered under IAI Guidance Note 11 (6)

- ix) The company wishes to introduce an '*added years gratuity*' option under the Flexible Benefit Scheme wherein the members can purchase additional years of gratuity. This benefit would be over and above the '*act gratuity*' as mentioned earlier. The additional years of gratuity will be calculated and paid separately with no gratuity ceiling involvement.

Discuss the actuarial basis and assumptions that you would use for determining the price of '*added years gratuity*'. (6)

- x) Discuss the problems associated with having the Gratuity Plan as being part of the '*Flexible Benefit Scheme*'. (4)

[50]

Solution 2 :-

i) The concept of '*Flexible Benefit Scheme*'

Flexible benefit schemes recognize that employees' needs change over time by allowing them to adjust their package each year and also perhaps on significant events e.g. childbirth. The choices offered by flexible benefit schemes can help meet the different needs of employees in several ways:

- for example allowing them to select from a menu of different benefit options, eg different accrual rates

It could be structured as:

- The company will have a bouquet of benefits covering all mandatory as well as other benefits – including Provident Fund, Gratuity, Leave, Life Insurance, Health Insurance, etc.

- The benefits will be categorized as CORE (which are compulsory) and OPTIONAL (which is choice based)
- The employee has to compulsorily choose all the CORE benefits and optionally choose some or all of the OPTIONAL benefits. In addition the employee can also choose the level of the OPTIONAL benefits.
- All statutory benefits will form part of CORE. The CORE will also include some other benefits at a minimum level – say Medical or Life Insurance.
- Each employee will have an '*Annual Benefit Amount*' to spend on their CORE and chosen OPTIONAL benefits.
- All benefits all have an individual price. The price of chosen benefits will be deducted from the '*Annual Benefit Amount*'.
- The employee can exchange one benefit for another.
- Some benefits may also have a maximum level and price.
- The employee can review and modify the flexible benefits every 1st of April.

(4 Marks)

ii)

Benefit	Reason	Benefit Level
Provident Fund	Statutory	As prescribed by Law
Gratuity	Statutory	As prescribed by Law
Leave	Mandatory as per Shops and establishment Act	20 days per annum (say)
Life Insurance	Family Protection	20 x Basic Salary (say)

(4 Marks)

iii) The factors would include:

- The type and level of benefits that would need to be priced for
- Whether the price has to be different for different categories of employees
- Whether the price is to be Fixed or as a Percentage of Salary or a combination of both
- The actual cost of providing such type and level of benefits.
- For some benefits, this could be the market cost of procuring the benefits from the market – say Life Insurance.
- For some benefits, this could be the actuarial value of the benefits – say Gratuity.
- For some benefits, this could be the accounting value of benefits.
- Ease of Administration and Operation and expenses involved
- Fairly Priced versus Cross Subsidies – especially if the employer wants to drive the employees to choose a particular benefit
- The frequency of review and modification of the price of benefits
- The frequency of review and modification of the choice of benefits by employees
- Is there any likely selection issue against the employer
- The perceived cost and benefit by the employees
- The competitor's benefit price levels in similar programs
- Comparison with the existing benefit structure
- The company's willingness and amount of investment

(6 Marks)

iv) Black's Model

Conceptual Framework

The Black's model, which provides a closed form solution for the value of a floor, assumes that the interest rate $R(k)$ follows a lognormal distribution with a specified volatility parameter. Based on this assumption, the value of the floorlet is given by the following equation:

$$V[\text{Floorlet}] = L * [t(k+1) - t(k)] * P[0, t(k+1)] * [R(F) * N(-d_2) - R(k) * N(-d_1)]$$

Where:

L = notional principal amount

$t(k+1)$ = time $(k+1)$

$t(k)$ = time k

$t(0)$ = valuation (balance sheet) date

$R(F)$ = guaranteed rate of interest

$R(k)$ = spread adjusted forward rate at time k

$d_1 = \{ \ln [R(k) / R(F)] + \sigma_k^2 * t(k) * 0.5 \}$ divided by $\sigma_k * t(k)^{0.5}$

$d_2 = d_1 - \sigma_k * t(k)^{0.5}$

σ_k = volatility parameter

$P[0, t(k+1)]$ = present value factor

$N(*)$ = cumulative normal probability values

Application of the Framework

The inputs and the process involved in applying the Black's Model for valuing the interest rate guarantee are as follows:

Inputs

The inputs required for using the Black's model are as follows:

- The Gilt Yield Curve [the zero coupon yield curve applicable to Government of India Bonds] as on the valuation date
- The investment return earned on the assets backing the PF Accumulation for five to ten years immediately preceding the valuation date
- The current guaranteed rate of return, which is typically equal to the rate of return declared by the Employees Provident Fund Organization [EPFO]
- The PF accumulation as on the Valuation Date
- The expected working life time of the members of the exempt provident fund as on the valuation date
- The demographic assumptions related to decrements such as future attrition rates and mortality rates
- The discount rate which is equal to the market yield on Government bonds [on the balance sheet date]. The term of the Government bonds must be equal to the decrement adjusted expected working life time of the employees.

Process

The following steps are involved in applying the Black's Model for valuing the interest rate guarantee embedded in an exempt provident fund:

- Obtain the continuously compounded Zero Coupon gilt yield curve [as on the balance sheet date] over the "decrement adjusted" expected working lifetime of the members of the exempt provident fund.
- Derive the one-year forward rates from the Zero-coupon yield curve obtained in the previous step
- Adjust the one-year forward rates for the yield spread between the portfolio rate of return and the yield on the gilts of an appropriate term. The portfolio rate of return refers to the rate of return on the asset portfolio backing the PF accumulation.
- Determine an appropriate volatility parameter for the spread adjusted one-year forward rates. This parameter can be estimated as the standard deviation of the historical rates of return on the asset portfolio backing the PF accumulation.
- Project the guaranteed rates of return based on the recent rate declared by the EPFO
- Use the Black's Model for estimating the value of the floorlet for each year of the decrement adjusted remaining working life time. The value of the floor will be equal to the sum of the values of the floorlets.
- The PVO [Present Value Obligation] of the Interest Guarantee is equal to the value of the floor.

(8 Marks)

v)
Employer

Gratuity Benefits: Paid to the Employees

- Deductible from income under section 37(1) of Income Tax Act, 1961. . It is because in terms of Section 17(1) (iii) of the Act, gratuity is a variant of salary and hence for the employer it is a necessary business expense.

Gratuity Fund: Contributions & Investment Income

Tax benefits are available to Gratuity Funds provided they are approved by the Commissioner of Income Tax under Part C of Fourth Schedule of Income Tax Act, 1961 and Part XIV of Income Tax Rules, 1962.

The conditions to be satisfied by a scheme to qualify for approval are set out in Rule 3 of the said Schedule. The main conditions include that the Scheme should be run by the Trustees, the trust should be irrevocable in nature, the employer should be contributor to the fund, funding of gratuity should be the sole purpose of the fund, contributions should not exceed 8.33% of salary per annum etc.

An approved Gratuity Fund is entitled to following Tax reliefs:

- As per Section 36(1) (V) of IT Act contributions paid by the employer are treated as business expenses of management.
- As per Central Board of Direct Taxes (CBDT), if the past service liability is funded by the employer in a single installment or in maximum five installments, the whole of the

contribution made by the employer will be treated as business expense in the year of payment.

- Income (whether interest or capital gains) out of the investments of the Trust is not taxable as per Section 10(25) (iv) of IT Act.

Employee

Gratuity Benefits: Received from the Employer

- Taxable under Section 15 of the Income Tax Act, 1961 as a variant of salary.
- However as per Section 10(10)(ii) of the IT Act, any gratuity received under the Gratuity Act is not taxable.
- Also, as per Section 10(10) (iii) of IT Act any gratuity which does not exceed one half month salary for each year of completed service, subject to such limit as notified by the Central Government is not taxable.

Gratuity Fund: Contribution by the Employer

The contributions (or premium in case of an insured Scheme) will not be treated as income in the hands of the employees and hence will not be taxed in their hands. This is as per a directive by the CBDT in which a ruling was given that the premium or the amounts payable by the Trustees of an approved Gratuity Fund to a life insurer or life insurers for obtaining a Group Gratuity Scheme will not be regarded as taxable perquisites under Section 17(2) (V) of Income-tax Act in the hands of the employees.

(4 Marks)

vi)

The factors would include:

- The current company contributions take account of the surplus of the fund and hence are lower than the perceived actuarial cost.
- The difference is temporary and would depend on the method and timing of amortization of the surplus.
- The company contributions would increase after the surplus is completely utilized and make the option unpopular among the employees.
- The current funding hence would not provide stability in the price and hence would not be a suitable choice for the price.
- The company needs to be careful to not adjust the actual contributions paid into the scheme once the surplus runs off as this would affect the security of the benefits.
- Alternatively the Standard Contribution Rate could be a stable choice as it would not take account of the surplus and closer to the perceived actuarial cost of the scheme.
- However the Standard Contribution Rate approach may not suit – as this is a global rate across all ages and sex. The young will subsidize the old and the males subsidize the females.
- The contribution rate should also be suitable under all circumstances; hence the method used in arriving at the current company contributions would be critical. For example, Projected Unit Method would not produce a stable rate if the average age/sex/salary distribution changed – say for a scheme closed to new entrants.
- The contribution rate should also be suitable under all circumstances; hence the assumptions used in arriving at the current company contributions would be critical. For example the changing demographics of scheme due to a large influx of new

entrants, say from a takeover.

- If there is an assumption for withdrawal, it may penalize the stayers and vice versa. Similar penalization happens for survivors as there is an assumption for mortality.
- The assumptions for funding would include an element of prudence and hence would be higher than required.
- The employees may not take up the option if the price is perceived to be higher than required which in the long term is not good.

(6 Marks)

vii)

The alternative methods could be:

- Age Related Rates
 - For stability, could be based on Attained Age Method and Age at Entry
 - Rates will be higher at younger ages, and hence would result in poor take up rate
- Sex Specific Rates
 - Would eliminate the cross subsidy between males and females
 - However would create discrimination issues
- Current Unit Method
 - Current Unit Method could be used to reduce the problems of withdrawal profits
- Accounting Cost
 - Accounting cost (accruals only) could be used for smoother realistic cost

(2 Marks)

viii)

Choice of Actuarial Assumptions

Purpose and basis:

The purpose for which the investigation is made has a bearing on the choice of assumptions. It is permissible to choose one set of assumptions for, say estimates leading to setting up of book reserves and disclosure in company accounts but another set of assumptions, for decision on contribution rates. Whereas for the purpose of funding the actuary may wish to be conservative in the choice of assumptions in the case of determining accrual of cost of for transfer values the assumptions should be realistic.

Special attention is to be given to those assumptions to which the Actuarial Liability and Contribution Rate are especially sensitive. The assumption chosen should be realistic and based on current experience. Whether it is a pension scheme or a gratuity scheme, or leave encashment scheme, the calculations of liability, contribution and cost estimates will be highly sensitive to assumptions relating to withdrawal rates, salary escalation and discounting rate. Relatively, mortality in service will be less important, though not mortality post-retirement in the case of pension schemes. Assumptions relating to salary escalation and discounting rate both involve an implicit assumption about rate of inflation. The past experience of the 'fund' relating to 'fund' interest earnings and salary escalation will also necessarily reflect to some extent the rate of inflation experienced in the past., Any attempt at assuming a future rate of inflation over a long period (as necessary for pensions and gratuity calculations) is a difficult task and any such assumption can therefore be no more than a hypothesis. The saving feature is that so long as the inflation

assumption implied in salary escalation and implied in discounting rate to be used are consistent, the liability and contribution calculations will be relatively less sensitive to changes in these two assumptions.

In the case of new schemes or where the question of choice of assumptions comes up for the first time, the actuary will have to use his experience of other schemes, the macro economic situation, currently accepted actuarial opinion and his judgement having regard to factors such as group size, type of employer's business, nature of employees' occupation and maintain a certain amount of consistency in the choice of assumptions in his work.

Rate of return assumption scenarios and basis:

In deciding the rate of interest to be assumed, the actuary has to take into account any statutory prescribed pattern of investment if applicable to the fund, the nature, term and yields under such prescribed investment, the likely delays and time lags in obtaining income from investments and the need of reinvestment of funds to match future liabilities and the interest rates at which such reinvestments are likely. The actuary should keep in mind that he is looking at a long term rate and take into account the broader economic scenario.

In the case of funds which are in existence, current experience, possible changes including restructuring of investments to new investment rates or otherwise has to be considered before deciding the rate of interest reflecting factors as enumerated above.

In the case of valuation for setting up book reserves and disclosure in accounts, the interest basis may be chosen having regard to the consideration in Para 7.4 above or based on the long term borrowing interest rates less the standard rate of tax on corporates. However, if circumstances warrant, a different rate of interest may be used with suitable justification being recorded.

Salary increase assumption and basis:

In projecting the pay increases it is necessary to keep in mind two factors -one is inflation level leading to a general change in salary level. The other is career progression of the members in their organisation. The actuary may combine the two and arrive at suitable salary growth levels. Or, if the actuary so desires, he may adopt a suitable salary scale and provide separately for the inflation factor.

Mortality assumption and basis:

The LIC (2006-08) modified Ultimate is the latest published mortality table available on Indian Assured Lives. This mortality table can be adopted by the actuary for making a suitable Guidance Note (GN) 11 assumption in respect of employee's mortality in service or period of deferment. However, the actuary has to examine suitability of these rates and where necessary a basis with an adjustment to the table by a rating up/down or by a mathematical formula may be used. It is open to him to use any other published mortality table which he feels is appropriate.

Under retirement benefit arrangements, where on retirement, the benefit amount is bought out by the company from a life or a pension office, an assumption in respect of mortality-in-retirement will not be required as simply the buy-out terms offered by the life or the pension office can be used for the purposes of quantifying the liabilities. However, in all other cases, for instance, when liabilities of an unfunded arrangement are being quantified or where the benefits on retirement are being directly disbursed by the

company or the trust, an assumption in respect of mortality-in-retirement will be required. we have LIC 1996-98 modified ultimate

It has be appreciated that the actual experience of mortality of the members of an employer-sponsored retirement benefit scheme may not replicate the experience of any published mortality table for a variety of socio-economic factors. It is therefore recommended that for sufficiently large schemes say of 20,000 members or more, the actuary may analyze the past mortality experience of the scheme provided sufficient and comprehensive data is available and based on the results of such an analysis adopt a suitable assumption for the future. For smaller schemes, where due to small size, statistically credible estimates are not possible, the actuary can rely more on published tables with an appropriate rating, if desired.

Attrition assumption and basis:

Withdrawal rates, both at early durations of service and near retirement date, have not only a significant impact on estimates of 'liability' and 'contributions' (more than of mortality in service) but are most difficult to estimate. The past may not be a guide to the future. Even if the past experience can be statistically analysed and produce some meaningful rates, the future experience of withdrawals will depend on general economic conditions as also the particular conditions affecting the given employer's business. Furthermore, withdrawal rates differ significantly from scheme to scheme and within a scheme from year to year. The only published information available in India is 'Withdrawal rates of employees covered under Insured group schemes'. NIA, December 1986. Group size, Public/Private Sector, Zone and Administrative Staff/Workers have been brought out in this investigation as major factors. The actuary has to examine these rates and any other information available to him and use his best possible judgment to cater to the long term nature of the actuarial estimates he is carrying out.

(6 Marks)

ix)

- There is neither any compulsion to pay for the choice every year nor any compulsion to pay at the same price level every year. Hence it is best to set up the price as a single premium which gives suitable credit for that premium.
- The arrangement as above may be difficult to administer given the number of employees and history of various years of flexible choices made.
- As there is cross subsidy between various categories of employees, the attitude of the employer in relation to this is important in determining an appropriate basis.
- The basis needs to be realistic so that it offers fair value for money and reduces cross subsidy to the extent possible.
- The basis should if any err on the prudent side – to protect against some selection against the employer. The funding basis may be suitable in this regard.
- The discount rate should take account of the return on scheme assets – and hence in this case the return used by the insurance company.
- The salary growth should be consistent with the assumed discount rate and the prevailing inflation. It should also take into account the company policy and the actual increases in the past couple of years, budgeted increases in the future, industry trends and macroeconomic scenario
- The leaver assumption should take into account the current trend and capture the long term trend.

(6 Marks)

x) Discuss the problems associated with having the Gratuity Plan as being part of the *Flexible Benefit Scheme*.

The problems include:

- The price could be high as is usual for a Defined Benefit Plan, and hence employees may not show interest in taking up.
- For those taking up, they would miss other benefits which are cheaper and may be more essential.
- Cross subsidies exist as is mentioned earlier – young / old; males / females; leavers / stayers etc.
- Administration would be complex and may involve higher expenses.
- The employees may not be able to plan well if the price is not stable and changes year on year.
- Not many companies may be offering this and hence difficult to gauge pricing and administration experience.
- The employees may not understand the benefit option and the corresponding price. They would tend to relate the contributions paid to the additional benefits and there would be issues with managing expectations
- The understanding and communication may be further difficult as there is an element of gratuity ceiling.
- It could be difficult to calculate the price for the benefit option without cross subsidies.
- There could be selection against the company.
- There would be additional complexities if the act is amended.
- Where the added gratuity exceeds the tax allowable payment limit, then the same would be taxable and would be viewed unfavorably by employees

(4 Marks)

[Total Marks-50]
